

CS-600: Software Design and Development

7-2 Activity: API Integration and Testing

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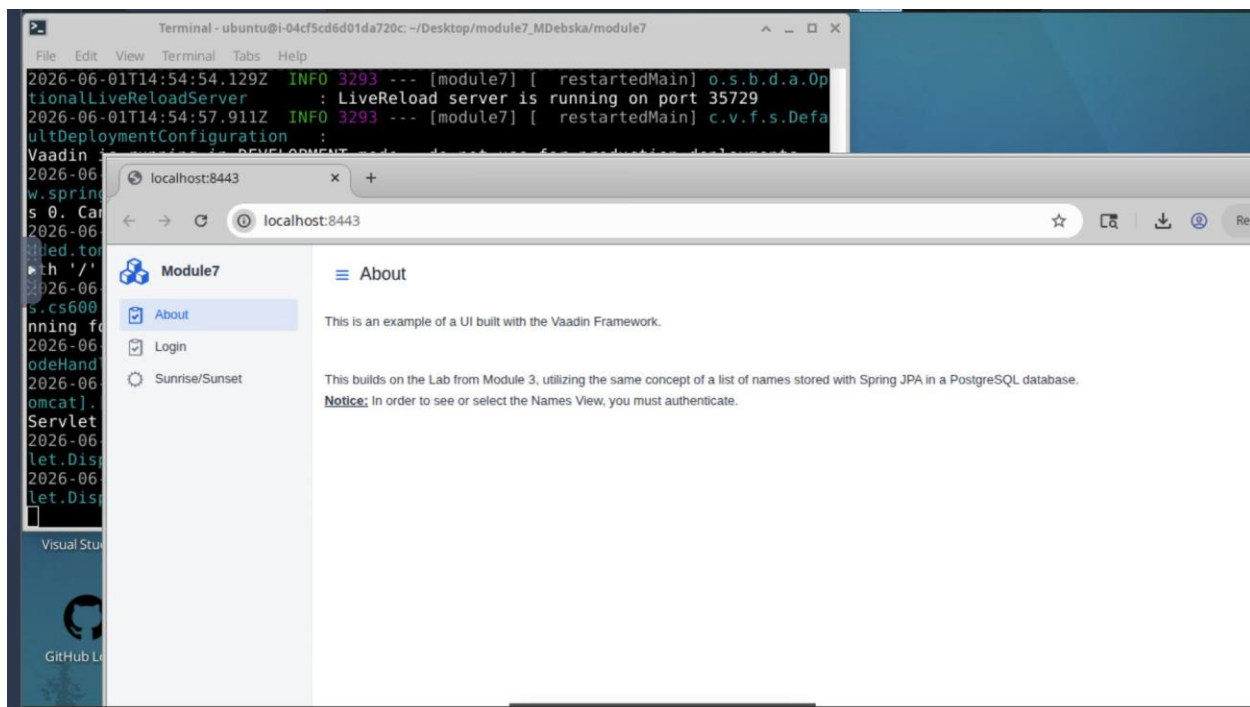
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API Integration and Testing Summary

For this activity, I integrated the Sunrise/Sunset API into the existing application. The application was configured to communicate with the external service by sending requests that included the selected location coordinates and date. The API response was processed, and the sunrise and sunset information were displayed within the user interface. To improve reliability, error-handling logic was implemented to manage communication failures and invalid responses. This ensures that users receive meaningful feedback if the API is unavailable or if an unexpected error occurs.

Testing and Verification

The integration was tested using the provided Maven testing framework. All tests completed successfully and returned a BUILD SUCCESS result, confirming that the application communicates correctly with the external API and processes responses as expected. More verification was performed by running the application and retrieving sunrise and sunset data for the Lincoln Memorial location. The application successfully displayed the location name, date, sunrise time, and sunset time obtained from the Sunrise/Sunset API. This confirmed that the integration was functioning correctly and that data exchange between the application and the external system was successful. The API integration met all project requirements by successfully connecting to the Sunrise/Sunset API, processing external data, implementing error handling, and passing all tests. The completed integration provides reliable communication between systems and provides correct sunrise and sunset information to users.



```
Terminal - ubuntu@i-04cf5cd6d01da720c: ~/Desktop/module7_MDebska/module7
File Edit View Terminal Tabs Help
ry.java:384) ~[spring-boot-3.5.6.jar:3.5.6]
    at org.springframework.boot.web.embedded.tomcat.TomcatServletWebServerFactory.customizeConnector(TomcatServletWebServerFactory.java:360) ~[spring-boot-3.5.6.jar:3.5.6]
    at org.springframework.boot.web.embedded.tomcat.TomcatServletWebServerFactory.getWebServer(TomcatServletWebServerFactory.java:209) ~[spring-boot-3.5.6.jar:3.5.6]
    at org.springframework.boot.web.servlet.context.ServletWebServerApplicationContext.createWebServer(ServletWebServerApplicationContext.java:193) ~[spring-boot-3.5.6.jar:3.5.6]
    at org.springframework.boot.web.servlet.context.ServletWebServerApplicationContext.onRefresh(ServletWebServerApplicationContext.java:167) ~[spring-boot-3.5.6.jar:3.5.6]
    ... 11 common frames omitted
Caused by: java.lang.IllegalStateException: Could not load store from 'keystore.p12'
    at org.springframework.boot.ssl.jks.JksSslStoreBundle.loadKeyStore(JksSslStoreBundle.java:140) ~[spring-boot-3.5.6.jar:3.5.6]
    at org.springframework.boot.ssl.jks.JksSslStoreBundle.createKeyStore(JksSslStoreBundle.java:107) ~[spring-boot-3.5.6.jar:3.5.6]
    ... 23 common frames omitted
Caused by: java.io.FileNotFoundException: keystore.p12
    at org.springframework.core.io.FileSystemResource.getInputStream(FileSystemResource.java:196) ~[spring-core-6.2.11.jar:6.2.11]
    at org.springframework.boot.ssl.jks.JksSslStoreBundle.loadKeyStore(JksSslStoreBundle.java:135) ~[spring-boot-3.5.6.jar:3.5.6]
    ... 24 common frames omitted

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 14.960 s
[INFO] Finished at: 2026-06-01T14:51:57Z
[INFO] -----
ubuntu@i-04cf5cd6d01da720c:~/Desktop/module7_MDebska/module7$ mvn spring-boot:run -Dspring-boot.run.arguments="--server.ssl.enabled=false"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.snhu.msps.cs600.module7 >-----
```

